



Foundations of Algebra: From Equations to Abstraction

A historical, mathematical, and philosophical study of structure, symmetry, and generality (32 weeks / 8 months).

Phase I – Linear Algebra: Structure and Transformation (Weeks 1–8)

Week	Mathematical Focus	Core Texts	Historical & Philosophical Readings	Guiding Question
1	Systems of equations, symbolic algebra	Axler Ch. 1; Strang Ch. 1–2	Viète <i>In Artem Analyticem Isagoge</i> (1591); Descartes <i>La Géométrie</i> (1637); Kline <i>Mathematical Thought from Ancient to Modern Times</i> Ch. 18	How did algebra emerge from the art of solving equations?
2	Vector spaces and linear combinations	Axler Ch. 2; Strang Ch. 3	Corry <i>Modern Algebra and the Rise of Mathematical Structures</i> Ch. 1; Cassirer <i>Substance and Function</i> Ch. 1	How did geometry become algebraized?
3	Basis, dimension, and subspaces	Axler Ch. 3–4; Hoffman & Kunze §1.4–1.6	Grassmann <i>Ausdehnungslehre</i> (1844); Crowe <i>A History of Vector Analysis</i> Ch. 2	How was “space” abstracted beyond geometry?
4	Linear maps and matrices	Axler Ch. 5–6; Strang Ch. 5	Cayley “A Memoir on the Theory of Matrices” (1858); Stillwell <i>Mathematics and Its History</i> Ch. 18	When did algebra become a theory of transformations?
5	Eigenvalues, diagonalization, invariants	Axler Ch. 7–8; Strang Ch. 6	Hamilton <i>Elements of Quaternions</i> (1866); Weyl <i>Philosophy of Mathematics and Natural Science</i> Ch. 3	What does an eigenvalue reveal about transformation and symmetry?
6	Inner products and orthogonality	Axler Ch. 9; Hoffman & Kunze §6.1–6.4	Dirichlet, “Über die Darstellung ganzer Zahlen”; Cassirer <i>Substance and Function</i> Ch. 4	How did orthogonality generalize geometric intuition?
7	Spectral theorem and quadratic forms	Axler Ch. 10–11	Hilbert <i>Integral Equations</i> (1904); Dieudonné <i>Historical Development of Algebraic Ideas</i>	How did infinite-dimensional methods transform algebra?
8	Applications and synthesis	Strang Ch. 7–8 (applications); review	Klein <i>Erlangen Program</i> (1872); Lakatos <i>Proofs and Refutations</i> §§	How did the concept of structure unify geometry and algebra?

Phase II – Abstract Algebra I: Groups and Symmetry (Weeks 9–16)

Week	Mathematical Focus	Core Texts	Historical & Philosophical Readings	Guiding Question
9	Groups and transformations	Pinter Ch. 1–5; D&F Ch. 1–2	Klein <i>Lectures on the Icosahedron</i> (1876); Wussing <i>Genesis of the Abstract Group Concept</i> Ch. 1	How did symmetry become a mathematical object?
10	Subgroups, cosets, and Lagrange’s theorem	Pinter Ch. 6–7; D&F §3.1–3.3	Lagrange (1770); Corry <i>Modern Algebra</i> Ch. 2	What is the algebraic meaning of structure?
11	Homomorphisms, quotient groups, normality	Pinter Ch. 8–10; D&F §3.4–3.5	Galois (1832); Corry <i>Modern Algebra</i> Ch. 3; Cassirer <i>Substance and Function</i> §§	How does structure preserve equivalence and transformation?
12	Rings and ideals	Pinter Ch. 13–15; D&F Ch. 7	Dedekind “Ideale”; Noether “Idealtheorie in Ringbereichen” (1921)	How does algebra capture divisibility and structure?
13	Fields, polynomials, and Gauss’s legacy	Pinter Ch. 16–17; D&F Ch. 13	Gauss <i>Disquisitiones Arithmeticae</i> §§I–III; Kline Ch. 25	What is the field concept, and why was it revolutionary?
14	Extensions and constructions	Pinter Ch. 18–19; D&F §13.4–13.6	Kronecker (1882); van der Waerden <i>Moderne Algebra</i> (1930)	How can algebra create new number systems?
15	Galois theory	Pinter Ch. 20; D&F Ch. 14; Artin <i>Galois Theory</i>	Galois (1832); Corry Ch. 4; Edwards <i>Galois Theory</i>	Why are some equations unsolvable by radicals?
16	Review & structural synthesis	Pinter Ch. 21; review	Noether <i>Invariante Variationsprobleme</i> (1918); Mac Lane <i>Mathematics: Form and Function</i> §§	What is “structure” in modern mathematics?

Phase III – Abstract Algebra II: Modern Structure and Representation (Weeks 17–24)

Week	Mathematical Focus	Core Texts	Historical & Philosophical Readings	Guiding Question
17	Modules and linear structures	Lang Ch. III §1–4	Noether (1921); Dieudonné <i>Development of Algebraic Ideas</i>	How do modules generalize vector spaces?
18	Exact sequences and homomorphisms	Lang Ch. III §5; Artin Ch. 4	Eilenberg & Mac Lane “General Theory of Natural Equivalences” (1945); Corry Ch. 5	How do homomorphisms encode relationships between structures?
19	Tensor products and bilinear maps	Lang Ch. X §1–3	Bourbaki <i>Elements of Mathematics</i> Vol. I; Cassirer	How does algebra handle multi-relational structure?

Week	Mathematical Focus	Core Texts	Historical & Philosophical Readings	Guiding Question
			<i>Philosophy of Symbolic Forms</i> Ch. 5	
20	Representation theory I	Fulton & Harris Ch. 1–2	Frobenius (1896); Weyl <i>Theory of Groups and Quantum Mechanics</i>	How can algebra represent symmetry?
21	Representation theory II	Fulton & Harris Ch. 3–4	Schur (1905); Weyl <i>Symmetry</i> (1952)	How does representation unify algebra and geometry?
22	Field extensions and algebraic closures	Lang Ch. V §1–3	Steinitz “Algebraische Theorie der Körper” (1910); Corry Ch. 6	When is algebra complete?
23	Algebraic numbers and rings of integers	Lang Ch. VI §1–2	Dedekind (1879); Hecke (1923)	How did number theory become structural algebra?
24	Bridge to category theory	Lang Ch. I §1–2; Mac Lane Ch. 1–2	Cassirer <i>Substance and Function</i> §§; Mac Lane <i>Categories for the Working Mathematician</i>	What is the logic of structure itself?

Phase IV – Unification and Applications (Weeks 25–32)

Week	Mathematical Focus	Core Texts	Historical & Philosophical Readings	Guiding Question
25	Categories and functors	Mac Lane Ch. 1–2	Lawvere (1963); Eilenberg–Mac Lane (1945)	How does category theory capture the essence of structure?
26	Natural transformations and universality	Mac Lane Ch. 3	Lawvere <i>Functorial Semantics</i> ; Ferreirós <i>Mathematical Knowledge and the Interplay of Practices</i>	What does universality mean in modern algebra?
27	Tensor categories and representations	Fulton & Harris Ch. 5	Tannaka, Grothendieck <i>Éléments de géométrie algébrique</i>	How do algebra and geometry become dual?
28	Algebra in analysis	Axler review; Kreyszig §Linear Operators	Hilbert (1904); von Neumann (1932)	Where do algebra and analysis meet?
29	Lie algebras and continuous symmetry	Hall Ch. 1–3	Lie (1870s); Cartan (1894); Corry Ch. 7	How did continuous symmetry reshape algebra?
30	Algebra in physics	Sakurai Ch. 1–2; Dirac <i>Principles of Quantum Mechanics</i>	Wigner (1939); Weyl <i>Philosophy of Mathematics and Natural Science</i>	How does algebra encode physical law?
31	Commutative algebra and geometry	Atiyah & Macdonald Ch. 1–2	Zariski (1947); Grothendieck (1960); Eppele (2006)	How do algebra and geometry mirror each other?

Week	Mathematical Focus	Core Texts	Historical & Philosophical Readings	Guiding Question
32	Synthesis and reflection	Review all texts	Cassirer <i>Substance and Function</i> Ch. 6; Lakatos <i>Proofs and Refutations</i>	What is the philosophical meaning of algebraic structure?

Perfect — that’s exactly the right instinct again.

Your algebra sequence (especially now that it runs 8 months) can be vastly enriched by embedding **historical and philosophical readings** that illuminate *how algebra became structural*, *how abstraction developed*, and *why symmetry, invariance, and formalism* became central to mathematics.

Below is the **revised, fully contextualized version** of your **Algebra syllabus bibliography**, carefully expanded to align historically and philosophically with each phase — from the symbolic revolution of Vieta and Descartes to the structural revolutions of Noether and Mac Lane.



Expanded Bibliography — Foundations of Algebra: From Equations to Abstraction

I. Core Mathematical Texts

Linear Algebra

1. **Axler, Sheldon.** *Linear Algebra Done Right*. 4th ed. Springer, 2024.
2. **Strang, Gilbert.** *Introduction to Linear Algebra*. 6th ed. Wellesley–Cambridge Press, 2022.
3. **Hoffman, Kenneth, and Ray Kunze.** *Linear Algebra*. 2nd ed. Pearson, 1971.
4. **Lang, Serge.** *Linear Algebra*. 3rd ed. Springer, 1987.
5. **Halmos, Paul R.** *Finite-Dimensional Vector Spaces*. Springer, 1974.
6. **Friedberg, Insel, and Spence.** *Linear Algebra*. 5th ed. Pearson, 2021.

Abstract and Modern Algebra

7. **Pinter, Charles.** *A Book of Abstract Algebra*. 2nd ed. Dover, 2010.
8. **Dummit, David S., and Richard M. Foote.** *Abstract Algebra*. 4th ed. Wiley, 2021.
9. **Artin, Michael.** *Galois Theory*. 2nd ed. Dover, 1998.
10. **Artin, Michael.** *Algebra*. 2nd ed. Pearson, 2010.
11. **Lang, Serge.** *Algebra*. Rev. 3rd ed. Springer, 2002.
12. **Jacobson, Nathan.** *Basic Algebra I & II*. 2 vols. Dover, 2009.
13. **Hungerford, Thomas.** *Algebra*. Springer, 1974.

14. **Atiyah, M. F., and I. G. Macdonald.** *Introduction to Commutative Algebra*. Addison–Wesley, 1969.
 15. **Fulton, William, and Joe Harris.** *Representation Theory: A First Course*. Springer, 1991.
 16. **Hall, Brian.** *Lie Groups, Lie Algebras, and Representations*. Springer, 2015.
 17. **Mac Lane, Saunders.** *Categories for the Working Mathematician*. 2nd ed. Springer, 1998.
 18. **Rotman, Joseph.** *Advanced Modern Algebra*. 3rd ed. AMS, 2010.
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II. Primary Historical Sources

Foundations of Algebraic Symbolism

1. **François Viète.** *In Artem Analyticem Isagoge*. Tours, 1591.
2. **René Descartes.** *La Géométrie*. Paris, 1637.
3. **Isaac Newton.** *Arithmetica Universalis*. Cambridge, 1707.
4. **Gottfried Wilhelm Leibniz.** “Nova Methodus pro Maximis et Minimis.” *Acta Eruditorum* (1684).
5. **Leonhard Euler.** *Elements of Algebra*. St. Petersburg, 1765.

Emergence of Modern Structure

6. **Joseph-Louis Lagrange.** “Réflexions sur la résolution algébrique des équations.” *Oeuvres complètes*, 1770.
7. **Niels Henrik Abel.** “Mémoire sur les équations algébriques.” 1824.
8. **Évariste Galois.** *Mémoire sur les conditions de résolubilité des équations*. 1832.
9. **Carl Friedrich Gauss.** *Disquisitiones Arithmeticae*. Leipzig, 1801.
10. **Arthur Cayley.** “A Memoir on the Theory of Matrices.” *Philosophical Transactions of the Royal Society* (1858).
11. **Hermann Grassmann.** *Die Lineale Ausdehnungslehre*. Leipzig, 1844.
12. **William Rowan Hamilton.** *Elements of Quaternions*. Dublin, 1866.
13. **Leopold Kronecker.** “Über den algebraischen Zahlkörper.” *Monatsberichte der Berliner Akademie*, 1882.
14. **Richard Dedekind.** “Über die Theorie der ganzen algebraischen Zahlen.” Braunschweig, 1879.
15. **Felix Klein.** *Vergleichende Betrachtungen über neuere geometrische Forschungen (Erlangen Program)*. 1872.
16. **David Hilbert.** *Grundlagen der Geometrie*. Leipzig, 1899.
17. **Ernst Steinitz.** “Algebraische Theorie der Körper.” *J. Reine Angew. Math.* 137 (1910).

18. **Emmy Noether.** “Idealtheorie in Ringbereichen.” *Mathematische Annalen* 83 (1921): 24–66.
 19. **B. L. van der Waerden.** *Moderne Algebra*. Berlin, 1930.
 20. **Saunders Mac Lane and Samuel Eilenberg.** “General Theory of Natural Equivalences.” *Trans. AMS* 58 (1945).
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III. Historical and Interpretive Studies

Development of Algebraic Structure

1. **Corry, Leo.** *Modern Algebra and the Rise of Mathematical Structures*. 2nd ed. Birkhäuser, 2004.
 2. **Wussing, Hans.** *The Genesis of the Abstract Group Concept*. Cambridge, MA: MIT Press, 1984.
 3. **Kleiner, Israel.** *A History of Abstract Algebra*. Birkhäuser, 2007.
 4. **Crowe, Michael J.** *A History of Vector Analysis*. University of Notre Dame Press, 1967.
 5. **Stillwell, John.** *Mathematics and Its History*. 4th ed. Springer, 2022.
 6. **Dieudonné, Jean.** *The Historical Development of Algebraic Ideas*. AMS, 1989.
 7. **Epple, Moritz.** “Geometric Concepts and the Structure of Modern Algebra.” In *The Architecture of Modern Mathematics*, eds. Ferreirós & Gray. Oxford, 2006.
 8. **Burton, David.** *The History of Mathematics: An Introduction*. McGraw-Hill, 2011.
 9. **Tobies, Renate.** *Emmy Noether: Mathematician Extraordinary*. Springer, 2019.
 10. **Bottazzini, Umberto, and Jeremy Gray.** *Hidden Harmony, Geometric Fantasies: The Rise of Complex Function Theory*. Springer, 2013.
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IV. Philosophy of Algebra, Structure, and Abstraction

Philosophy of Mathematics and Structuralism

1. **Cassirer, Ernst.** *Substance and Function*. Open Court, 1923.
2. **Cassirer, Ernst.** *The Philosophy of Symbolic Forms, Vol. 3: The Phenomenology of Knowledge*. Yale University Press, 1957.
3. **Weyl, Hermann.** *Philosophy of Mathematics and Natural Science*. Princeton University Press, 1949.
4. **Whitehead, Alfred North.** *Science and the Modern World*. Macmillan, 1925.
5. **Whitehead, Alfred North, and Bertrand Russell.** *Principia Mathematica*. 3 vols. Cambridge University Press, 1910–13.
6. **Hilbert, David.** *Foundations of Geometry*. Trans. E. J. Townsend. Open Court, 1902.

7. **Mac Lane, Saunders.** *Mathematics: Form and Function*. Springer, 1986.
 8. **Shapiro, Stewart.** *Thinking About Mathematics: The Philosophy of Mathematics*. Oxford University Press, 2000.
 9. **Corfield, David.** *Towards a Philosophy of Real Mathematics*. Cambridge University Press, 2003.
 10. **Awodey, Steve.** *Category Theory*. 2nd ed. Oxford University Press, 2010.
 11. **Ferreirós, José.** *Mathematical Knowledge and the Interplay of Practices*. Princeton University Press, 2016.
 12. **Lakatos, Imre.** *Proofs and Refutations*. Cambridge University Press, 1976.
 13. **Maddy, Penelope.** *Defending the Axioms*. Oxford University Press, 2011.
 14. **Feferman, Solomon.** *In the Light of Logic*. Oxford University Press, 1998.
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V. Modern Context and Interdisciplinary Extensions

Category Theory, Computation, and Structure

1. **Lawvere, F. William.** “Functorial Semantics of Algebraic Theories.” *PNAS* 50 (1963): 869–872.
2. **Lawvere, F. W., and Stephen Schanuel.** *Conceptual Mathematics*. Cambridge University Press, 2009.
3. **Turing, Alan.** “On Computable Numbers, with an Application to the Entscheidungsproblem.” *Proc. LMS* (1936).
4. **Kleene, Stephen.** *Mathematical Logic*. Wiley, 1967.
5. **Kunen, Kenneth.** *Set Theory*. College Publications, 2011.
6. **Mac Lane, Saunders, and Ieke Moerdijk.** *Sheaves in Geometry and Logic*. Springer, 1992.
7. **Awodey, Steve, and Hans Halvorson, eds.** *Philosophy of Mathematics: Structure and Ontology*. Oxford University Press, 2022.
8. **Corry, Leo, and Jeremy Gray, eds.** *The Architecture of Modern Mathematics*. Oxford University Press, 2006.
9. **Ewald, William, ed.** *From Kant to Hilbert: A Source Book in the Foundations of Mathematics*. Oxford University Press, 1996.